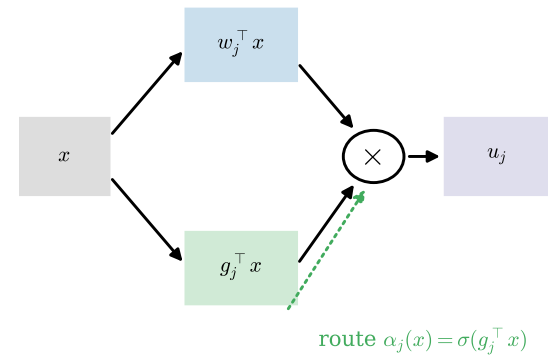
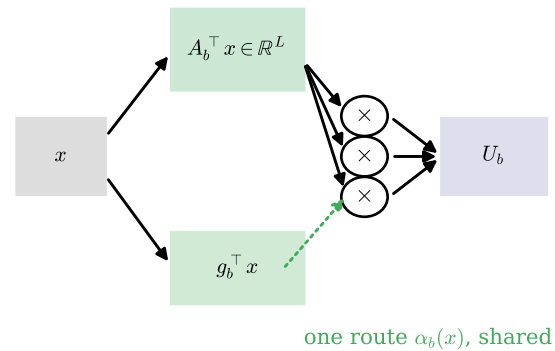


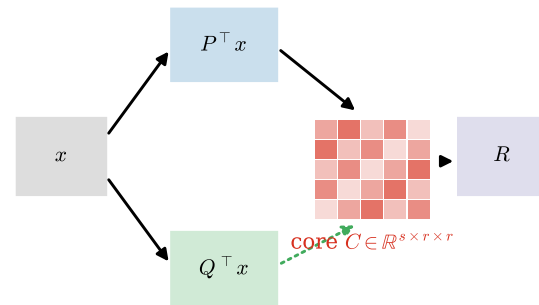
A) SwiGLU: routed CP

1 route $\rightarrow$ 1 rank-1 atom m routes, m atoms

B) LL1 / block-CP

1 route $\rightarrow$ 1 rank- L block B routes, BL atoms, $V_b = U_b A_b^\top$ 

C) dense Tucker

1 route $\rightarrow$ full-rank $V_j = RC^{(j)}$ r routes, all-to-all core, gauge freedomD) matched parameters ($d = 512$, $N = 2.29\text{M}$)